

1. 10 mL of liquid 'A' and 20 mL of liquid 'B' are mixed at 25°C. The volume of the solution was measured to be 30.1 mL then,
- (a) $\Delta H_{\text{mix}} > 0$, Solution shows negative deviation from Raoult's law.
(b) $\Delta H_{\text{mix}} < 0$, Solution shows negative deviation from Raoult's law.
(c) $\Delta H_{\text{mix}} > 0$, Solution shows positive deviation from Raoult's law.
(d) $\Delta H_{\text{mix}} < 0$, Solution shows positive deviation from Raoult's law.
2. The mixture which shows positive deviation from Raoult's Law is
- (a) n-hexane and n-heptane (b) Bromoethane and chloroethane
(c) **Ethanol and Acetone** (d) Chloroform and Acetone
3. A student took two glasses of pure water from a water filter. He cools glass-A in a fridge and warms the other glass - B on a stove. On comparing the solubility of oxygen in H_2O in glass-A and glass-B, he states that glass-'A' contains
- (a) **more oxygen than glass-B.** (b) less oxygen than glass-B.
(c) same amount of oxygen as in glass-B. (d) zero concentration of oxygen.
4. A binary solution has two components 'A' and 'B'. The mole fraction of component 'A' is 0.5, then the number of moles of components 'A' and 'B' in the solution is
- (a) $n_A < n_B$ (b) $n_A > n_B$ (c) **$n_A = n_B$** (d) All of these
5. The mass percentage (W/W) of glucose in water is 10% means.
- (a) 10g of glucose dissolved in 100 g of water (b) 10g of glucose dissolved in 90 mL of water
(c) 10g of glucose dissolved in 100 mL of water (d) **10g of glucose dissolved in 90 g of water**
6. **Statement-1:** If on mixing the two liquids, the solution becomes hot, it implies that it shows negative deviation from Raoult's law.
Statement-2: Solutions which show negative deviation are accompanied by decrease in volume.
- (a) **Both Statement-1 and Statement-2 are true.** (b) Statement-1 is true but Statement-2 is false;
(c) Both Statement-1 and Statement-2 are false. (d) Statement-1 is false but Statement-2 is true.
7. Van't Hoff factor (i) for a non-electrolyte in a solution is ____ **(one)**
8. Van't Hoff factor for KCl solution assuming the complete dissociation is ____ **(Two)**
9. The sum of mole fractions of all the components in a binary solution is equal to ____ **(one)**
10. Name the concentration term which is commonly used in medicine and pharmacy. Write the definition and mathematical equation for that concentration term.

Ans: mass by volume percentage **OR** $\left(\frac{m}{v}\right)\%$. **OR** $\left(\frac{w}{v}\right)\%$.

It is the mass of solute dissolved in 100 mL of the solution.

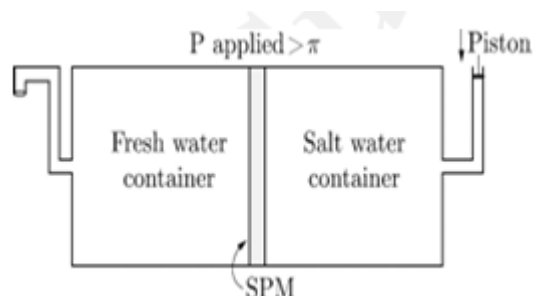
$$\% \text{ w/v} = \frac{\text{Mass of Solute (g)}}{\text{Volume of Solution (mL)}} \times 100$$

11. Write three reasons to justify that osmotic pressure method has the advantage over other colligative methods for the measurement of molar mass of the macromolecules.

Ans:

- i) It can be measured around the room temperature
- ii) Instead of molality, term Molarity of the solution is used
- iii) Its magnitude is large even for very dilute solutions.
- iv) It is useful for determination of molar mass of Biomolecules which are unstable at high temperature
- v) It is useful for determination of molar mass of poor soluble polymer

12. Given below is the sketch of a plant for carrying out a process.



- (i) Name the process occurring in the given plant.
- (ii) Name any one SPM which can be used in this plant.
- (iii) Give one practical use of the plant.

Ans: Reverse Osmosis

Ans: Cellopane membrane

Ans: De-salination of sea water

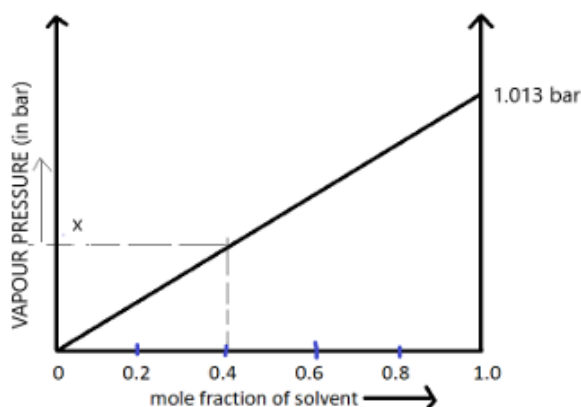
13. Henry's law constant for the molality of methane in benzene at 298 K is 4.27×10^5 mm Hg. Calculate the solubility of methane in benzene at 298 K under 760 mm Hg.

Ans: Given: $K_H = 4.27 \times 10^5$ mm Hg, $P = 760$ mmHg

$$p = K_H \times x_{\text{Methane}} \Rightarrow x_{\text{Methane}} = \frac{p}{K_H}$$

$$x_{\text{Methane}} = \frac{760 \text{ mmHg}}{4.27 \times 10^5 \text{ mm Hg}} \quad x_{\text{Methane}} = 1.78 \times 10^{-3}$$

14. If a solution obeys Raoult's law for all concentrations, its vapour pressure would vary linearly from zero to the vapour pressure of pure solvent as shown in graph. Calculate the value of 'x'.



Given: $x_1 = 0.4$, $p_1^0 = 1.013 \text{ bar}$

$$p_1 = p_1^0 x_1 \Rightarrow p_1 = 0.4 \times 1.013 \text{ bar} \Rightarrow p_1 = 0.4052 \text{ bar}$$

- 15.** Calculate the mass of a non-volatile solute which should be dissolved in 114g octane to reduce its vapour pressure to 80%. [Given Molecular formula of octane is C_8H_{18} ; Molar mass of solute = 40 g mol^{-1}].

Ans: $w_1 = 114 \text{ g}$, $w_2 = ?$ $p_1^0 = 100$, $p_1 = 80$

$$M_1 = 12 \times 8 + 1 \times 18 = 114 \text{ g mol}^{-1}, M_2 = 40 \text{ g mol}^{-1}$$

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{w_2 X M_1}{M_2 X w_1} \Rightarrow \frac{100 - 80}{100} = \frac{w_2 \times 114 \text{ g/mol}}{40 \frac{\text{g}}{\text{mol}} \times 114 \text{ g}} \Rightarrow 0.2 = \frac{w_2}{40} \Rightarrow w_2 = 8 \text{ g}$$

- 16.** The boiling point of benzene is 353.23K. Calculate the mass of non-volatile solute is to be added to 90g of benzene such that it boils at 354.11K. The molar mass of solute is 58 g mol^{-1} [Given: $K_b = 2.53 \text{ K Kg mol}^{-1}$].

Ans: $M_2 = 58 \text{ g mol}^{-1}$, $w_1 = 90 \text{ g}$,

$$T_b = 353.23 \text{ K} \quad T_b^0 = 354.11 \quad \Delta T_b = T_b - T_b^0 \Rightarrow T_b^0 = 354.11 \text{ K} - 353.23 \Rightarrow K = 0.88 \text{ K}$$

$$M_2 = \frac{1000 \times w_2 \times K_b}{\Delta T_b \times w_1} \Rightarrow w_2 = \frac{58 \frac{\text{g}}{\text{mol}} \times 0.88 \text{ K} \times 90 \text{ g}}{2.53 \text{ K Kg mol}^{-1} \times 1000} \Rightarrow w_2 = 1.81 \text{ g/mol}$$

- 17.** Calculate the osmotic pressure of 0.5%(w/v) aqueous solution sucrose at 300K. [$R = 0.083 \text{ L bar mol}^{-1}$ and molar mass of sucrose = 342 g mol^{-1}]

Ans: 0.5 g sucrose in 100 mL solution $\Rightarrow 5 \text{ g/L}$

$$w_2 = 5 \text{ g}, \quad V = 1 \text{ L} \quad T = 300 \text{ K}, \quad M_2 = 342 \text{ g mol}^{-1} \quad R = 0.083 \text{ L bar mol}^{-1} \text{ K}^{-1}$$

$$M_2 = \frac{w_2 R T}{\pi V} \quad \text{OR} \quad \pi = \frac{w_2 R T}{M_2 V}$$

$$\pi = \frac{5.0 \text{ g} \times 0.083 \text{ bar L K}^{-1} \text{ mol}^{-1} \times 300 \text{ K}}{342 \text{ g mol}^{-1} \times 1 \text{ L}} \Rightarrow \pi = 0.364 \text{ bar}$$

- 18.** Calculate the osmotic pressure in Pascals exerted by a solution prepared by dissolving 0.925 g of polymer of molar mass 185000 in 500 mL of water at 37°C [$R = 8.314 \times 10^3 \text{ Pa L K}^{-1} \text{ mol}^{-1}$]

Ans: $w_2 = 0.925 \text{ g}$ $V = 450 \text{ mL} = 0.5 \text{ L}$ $M_2 = 185000$ $T = 37^\circ \text{C} = 37^\circ + 273 = 310 \text{ K}$

$$R = 8.314 \times 10^3 \text{ Pa L mol}^{-1} \text{ K}^{-1}$$

$$\pi = \frac{w_2 R T}{M_2 V} \Rightarrow \pi = \frac{0.925 \text{ g} \times 8.314 \times 10^3 \text{ Pa L K}^{-1} \text{ mol}^{-1} \times 310 \text{ K}}{185000 \text{ mol}^{-1} \times 0.5 \text{ L}} \Rightarrow \pi = 25.77 \text{ Pa}$$

Few important and repeated questions:

- 1.** Calculate the mass percentage of benzene (C_6H_6) and carbon tetrachloride (CCl_4) if 22 g of benzene is dissolved in 122 g of carbon tetrachloride (NCERT)

Ans: Mass of Benzene = 22g mass of Carbon tetrachloride = 122 g

$$\text{Mass of solution} = \text{mass of benzene} + \text{mass of carbon tetrachloride} = 22 + 122 = 144 \text{ g}$$

$$\text{Mass \% of Benzene} = \frac{\text{Mass of benzene solution}}{\text{Total mass of the solution}} \times 100 \Rightarrow \frac{22 \text{ g}}{144 \text{ g}} \times 100 \Rightarrow 15.27\%$$

$$\text{Mass percentage of Carbo tetrachloride} = \frac{\text{Mass of carbon tetrachloride in the solution}}{\text{Total mass of the solution}} \times 100$$

$$\Rightarrow \frac{122g}{144g} \times 100 = 84.72\%$$

2. Molarity (M), molality (m) and mole fraction (χ) are some methods for expressing concentration of solutions. Which of these are temperature dependent? Give reason.

Ans: Molarity (M), it depends on volume, **volume depends on temperature**, Molarity $\propto \frac{1}{\text{Temperature}}$

3. How solubility of gas in liquid varies with (i) temperature and (ii) pressure?

Ans: Solubility **increases** with decrease in **temperature** and increase in **pressure**

4. State Henry's law. Write the mathematical expression of Henry's law

Ans: Mole fraction of gas in the solution is proportional to the partial pressure of the gas over the solution.

$$p \propto \chi \Rightarrow p = K_H \chi$$

5. Name the law behind the dissolution of CO₂ gas in soft drinks under high pressure

Ans: Henry's law

6. The vapour pressure of pure liquids 'A' and 'B' are 450 and 700 mm Hg respectively, at 350 K. Find out the composition of the liquid mixture if total vapour pressure is 600 mm Hg.

Ans: Given: $P_A^0 = 450 \text{ mmHg}$ $P_B^0 = 700 \text{ mmHg}$

$$P_{\text{Total}} = P_B^0 + (P_A^0 - P_B^0) \chi_A$$

$$600 \text{ mmHg} = 700 + (450 - 700) \chi_A$$

$$250 \chi_A = 100 \Rightarrow \chi_A = \frac{100}{250} \Rightarrow \chi_A = 0.40$$

$$\chi_A + \chi_B = 1 \Rightarrow \chi_B = 1 - 0.40 \Rightarrow \chi_B = 0.60$$

6. Give any two difference between ideal and non ideal solution

Ideal Solutions	Non ideal Solutions
They Obey raoult's law at all range of concentration	They do not Obey raoult's law at all range of concentration
$\Delta_{\text{mix}}H=0$	$\Delta_{\text{mix}}H \neq 0$
$\Delta_{\text{mix}}V=0$	$\Delta_{\text{mix}}V \neq 0$
If the intermolecular attractive forces between the A-A and B-B are nearly equal to those between A-B, Ex: benzene and toluene	Intermolecular attractive forces between A-A and B-B are weaker or stonger than those between A-B Ex:, ethanol+water, HNO₃ +water

7. Write any three difference between positive and negative deviation from Raoult's law

Positive Deviation From Raoult's law	Negative Deviation From Raoult's law
i) $\Delta_{\text{mix}}H > 0$	i) $\Delta_{\text{mix}}H < 0$
ii) $\Delta_{\text{mix}}V > 0$	ii) $\Delta_{\text{mix}}V < 0$

iii) A-B interaction is weaker than A-A & B-B	iii) A-B interaction is Stronger than A-A & B-B
Ex: Ethanol + water,	Ex: Water + nitric acid

8. What is azeotrope?? Give an example

Ans: binary mixtures having the same composition in liquid and vapour phase and boil at a constant temperature. **EX:** 95% ethanol + 5% water, 68% HNO₃ + 32% water

9. 5.8 gm of a non volatile solute was dissolved in 100g of carbon disulphide (molar mass=76). The vapour pressure of the solution was found to be 190mm Hg. Calculate the molecular mass of the solute. The vapour pressure of pure carbon disulphide(CS₂) 195mmHg (**A-16, S-18**)

Ans: $p_1^0 = 195\text{mmHg}$; $p_1 = 190\text{mmHg}$;

$M_1 = 76\text{ g mol}^{-1}$; $w_2 = 5.8\text{ g}$; $w_1 = 100\text{g}$, $M_2 = ?$

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{w_2 X M_1}{M_2 X w_1} \Rightarrow \frac{195\text{mmHg} - 190\text{mmHg}}{195\text{mmHg}} = \frac{5.8\text{g} \times 76\text{g mol}^{-1}}{M_2 \times 100\text{g}} \Rightarrow M_2 = 171.912\text{ g/mol}$$

10. Boiling point of water at 750 mm Hg is 99.63°C. How much sucrose is to be added to 500 g of water such that it boils at 100°C (**Ncert text**)

Ans: Given: $W_2 = ?$, $w_1 = 500\text{gm}$

$$T_b = 100^\circ\text{C}, \quad T_b^0 = 99.63^\circ\text{C} \Rightarrow \Delta T_b = T_b - T_b^0 \Rightarrow \Delta T_b = 100^\circ\text{C} - 99.63^\circ\text{C} \Rightarrow \Delta T_b = 0.37\text{K}$$

Molar mass of sucrose(C₁₂H₂₂O₁₁) $M_2 = (12 \times 12 + 1 \times 22 + 16 \times 11) = 342\text{ g/mol}$

$$M_2 = \frac{1000 \times w_2 \times K_b}{\Delta T_b \times w_1} \Rightarrow W_2 = \frac{\Delta T_b \times w_1 \times M_2}{K_b \times 1000}$$

$$W_2 = \frac{0.37\text{ K} \times 500\text{ g} \times 342\text{ g/mol}}{0.52\text{ K kg/mol} \times 1000} \Rightarrow W_2 = 121.6\text{ g}$$

11. A solution containing 18g of non volatile ,non electrolyte solute dissolved in 200g of water freezes at 272.07K. calculate the molecular mass of solute. Given: $K_f = 1.86\text{ K kg/mol}$, freezing point of water=273K)

Ans: $W_2 = 18\text{g}$, $w_1 = 200\text{g}$, $K_f = 1.86\text{ K kg mol}^{-1}$, $T_f = 272.0\text{K}$, $T_f^0 = 273\text{K} \Rightarrow \Delta T_f = T_f^0 - T_f \Rightarrow \Delta T_f = 273\text{K} - 272.07\text{K} \Rightarrow \Delta T_f = 0.93\text{K}$

$$\Delta T_f = \frac{1000 \times w_2 \times K_f}{M_2 \times w_1} \Rightarrow M_2 = \frac{1000 \times w_2 \times K_f}{\Delta T_f \times w_1} \Rightarrow M_2 = \frac{1000 \times 18\text{g} \times 1.86\text{ K kg/mol}}{0.93\text{K} \times 200\text{g}} \Rightarrow M_2 = 180\text{g/mol}$$

12. 450cm³ of an aqueous solution of a protein contains 1.0g of the protein. The osmotic pressure of such a solution at 310K is found to be $3.1 \times 10^{-4}\text{ bar}$. Calculate the molar mass of the protein. ($R = 0.083\text{L bar mol}^{-1}\text{ K}^{-1}$)

$$M_2 = \frac{w_2 R T}{\pi V} \Rightarrow M_2 = \frac{1.0\text{g} \times 0.083\text{L bar mol}^{-1}\text{K}^{-1} \times 310\text{K}}{3.1 \times 10^{-4}\text{ bar} \times 0.450\text{L}} \Rightarrow M_2 = 1,84,444\text{ g/mol}$$

13. Vapour pressure of water at 293 K is 17.535 mm Hg. Calculate the vapour pressure of water at 293 K when 25 g of glucose is dissolved in 450 g of water

Ans: Given: $p_1^0 = 17.535\text{ mmHg}$ $p_1 = ?$ $w_2 = 25\text{g}$ $w_1 = 450\text{ g}$

molar mass of glucose(C₆H₁₂O₆) $M_2 = 12 \times 6 + 1 \times 12 + 16 \times 6 = 180\text{ g/mol}$

Molar mass of water(H₂O) $M_1 = 1 \times 2 + 16 \times 1 = 18\text{ g/mol}$

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{w_2 X M_1}{M_2 X w_1} \Rightarrow \frac{17.535 - p_1}{17.535} = \frac{25\text{ g} \times 18\text{g mol}^{-1}}{180\text{ g mol}^{-1} \times 450\text{g}} \quad p_1 = 17.438\text{ mm}$$